



APEXA RAMI

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SUMMARY

Software developer with a strong foundation in Python, backend development, and cloud technologies. Experienced in building scalable systems, REST APIs, and working with AWS and CI/CD pipelines. Recently focused on machine learning, Generative AI, and MLOps, with hands-on experience in developing LLM-based workflows and integrating AI into real-world applications.

WORK EXPERIENCE

Research Intern – Generative AI (LLMs) Materialise, Belgium

October 2025 -March 2026

Worked in the Central Research and Technology unit, contributing to the development of Generative AI solutions using Large Language Models for enterprise use cases. Designed and implemented a Deep Research system capable of multi-step reasoning, iterative retrieval, and structured response generation, along with web search-enabled AI workflows to improve real-time information access. Developed backend services using FastAPI and integrated LLM APIs to support scalable AI-driven functionalities. Deployed and managed services on AWS and contributed to Azure-based MLOps pipelines for model integration and workflow automation. Additionally, worked on improving system architecture and enhancing the codebase to increase scalability, performance, and overall research efficiency while collaborating with cross-functional teams.

Software Developer, Syndell technologies

November 2022 - November 2023

Developed web applications and RESTful APIs using PHP, Node.js, and Java. Built responsive front-ends with HTML, CSS, JavaScript, and Bootstrap. Delivered e-commerce platforms with payment systems, order tracking, and supplier API integrations. Managed databases (PostgreSQL, MongoDB) and deployed applications via Docker, AWS/GCP, and CI/CD pipelines. Contributed in Agile teams through code reviews, sprint planning, and documentation.

Intern Web-designer & SEO expert, IT-HUB Software development

Mar 2021 - Dec 2021

Designed responsive websites, prioritizing UX. Implemented SEO strategies, boosting traffic. Conducted keyword research and content optimization. Used Google Analytics for performance tracking. Collaborated with clients to deliver solutions. Provided ongoing maintenance and support.

Intern in Home Automation Product manufacturing , Varni Digital

Feb 2020 - Jan 2021

Assisted in the development and testing of IoT-based home automation products using Python. Conducted market research to guide product improvements and collaborated with teams to address technical challenges. Participated in product demonstrations and feedback sessions, contributing to documentation and reporting. Gained practical skills in IoT, Python, and industry knowledge.

EDUCATION

Master of Engineering in Electrical and Microsystem

March 2024 - present

Ostbayerische Technische Hochschule (OTH) Regensburg

- Major Subjects: Electronics Production Engineering, Introduction to **AI & Machine Learning**, **Advanced Engineering Mathematics**, Deep learning, **OOP with Python**, Applied Optics, LED Technology, Applied Optoelectronics, Project Management, Advanced Semiconductor Technology.

Bachelor of Engineering in Electronics and communications

May 2018 - May 2022

Gujarat Technological University

- Specialized in Artificial Intelligence, IoT, and VLSI Design.
- Gained practical experience in AI algorithms, IoT protocols, and VLSI simulations.
- Strong foundation in C programming for embedded systems and algorithm development.
- Completed projects integrating AI and IoT for smart system applications.

SKILLS

- Programming & AI: Python, C++, SQL, LLMs, Generative AI, RAG, LangChain, AI Agents, Deep Research Systems, MCP Server Integration, NumPy, Pandas, REST APIs
- Backend: FastAPI, Node.js, MongoDB
- Cloud & MLOps: AWS (ECS), Azure OpenAI, Azure MLOps, Docker, CI/CD (GitLab, YAML)
- Scripting: Bash, YAML
- Tools: Git, MLflow, OpenCV
- Simulation: Siemens Tecnomatix Plant Simulation (Python integration)

PROJECTS

AI-Driven Mental Health Support Chatbot

Case-study

This project involves creating a web-based mental health support chatbot that utilizes artificial intelligence to provide personalized and empathetic assistance to users experiencing mental health challenges. The chatbot integrates natural language processing (NLP) algorithms to engage in meaningful conversations, offer empathetic responses, and provide valuable resources and support tailored to users' individual needs

Smart Grid Load Forecasting with MLOps

Case-study

- Technologies: Python, FastAPI, Docker, GitLab CI/CD, Azure, MLflow
- Built a machine learning model to forecast energy demand in smart grids.
- Developed a FastAPI-based API for real-time predictions.
- Automated model training and deployment using MLflow and GitLab CI/CD.
- Deployed the system with Docker on Azure for scalability.

Stock Price Prediction Using LSTM Neural Networks

Case-Study

- Technologies: Python, LSTM, Flask, Machine Learning
- Built a web app for predicting stock prices using an LSTM model.
- Analyzes historical data to forecast trends and provide investment recommendations.
- Users can select stocks and view predictions via a simple interface.
- Designed for extensibility, allowing future model and stock additions.

Contour Detection using Computer Vision

Mini Project

- Designed a system for object boundary detection, utilizing unsupervised learning techniques for image segmentation. This system demonstrates skills transferable to predictive maintenance tasks requiring precise pattern and anomaly detection in sensor data.
- Technologies Used: Python, OpenCV, Scikit-learn

Inventory Management System

Client Project

- Technologies: Python, Django, PostgreSQL, REST API, AWS
- Built a web-based inventory management system for real-time stock control.
- Integrated supplier APIs and implemented role-based authentication.
- Optimized database performance and deployed the application on AWS.

E-Commerce Web Application

Client Project

- Technologies: MongoDB, Express.js, React, Node.js, Stripe
- Created a full-stack e-commerce site with product listing, cart, and payment integration.
- Implemented user login and Redux-based state management.
- Improved speed and UX through responsive design and lazy loading.
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Website Performance Optimization (PageSpeed Improvement Project)

Client Project

- Technologies: React, Next.js, Webpack, Lighthouse, Google Cloud
- Boosted Google PageSpeed score from 61 to 95 by optimizing image delivery, minifying JS/CSS, and implementing server-side rendering.
- Reduced bounce rate by 30% and increased average session time through faster load times.
- Deployed optimized assets via CDN for global performance consistency.
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CI/CD Pipeline Automation (DevOps Efficiency Initiative)

Client Project

- Technologies: GitLab CI/CD, Docker, Kubernetes, AWS
- Automated build, test, and deployment workflows to reduce release time by 60%.
- Integrated static code analysis and unit tests into the pipeline for quality assurance.
- Enabled rollbacks and blue-green deployments to minimize downtime.

LANGUAGES

- English
- German(A2)